Suspicious Outbound Network Traffic Investigation

Wireshark • DNS • TCP/TLS Analysis • SOC Triage • Incident Escalation

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Case Overview
This simulated SOC investigation follows unusual outbound traffic from an internal workstation through DNS resolution, TCP connection establishment, TLS negotiation, repeated outbound-heavy transfers, and periodic beacon-like behavior. The exercise demonstrates Wireshark filtering, packet interpretation, conversation analysis, timeline reconstruction, evidence-based reasoning, uncertainty management, and Tier 1 escalation writing.

Objectives

- Isolate suspicious host traffic using Wireshark display filters.
- Interpret DNS, TCP, and TLS traffic during an investigation.
- Build a chronological network timeline and distinguish confirmed evidence from hypotheses.
- Recognize outbound-heavy transfers and periodic beacon-like communication.
- Produce concise Tier 1 escalation notes using Summary → Assessment → Recommendation.

Phase 1 — Initial Network Alert

Alert: Unusual outbound network activity detected from internal workstation 192.168.1.45. Severity: Medium.

```
3.441220  192.168.1.45 → 8.8.8.8        DNS      Standard query A
updates-check.test
 3.452810  8.8.8.8 → 192.168.1.45        DNS      Response A 198.51.100.42
 3.455901  192.168.1.45 → 198.51.100.42   TCP      52314 → 443 [SYN]
 3.463782  198.51.100.42 → 192.168.1.45   TCP      443 → 52314 [SYN, ACK]
 3.464103  192.168.1.45 → 198.51.100.42   TCP      52314 → 443 [ACK]
```

Display Filter

```
ip.addr == 192.168.1.45
```

My Investigation Response

The internal host 192.168.1.45 sent a DNS query for updates-check.test, which resolved to 198.51.100.42. The host then initiated a TCP connection to the resolved address on destination port 443. The SYN, SYN/ACK, and ACK sequence completed successfully, confirming that a TCP connection was established. At this stage, the evidence does not prove that application data was exchanged or that the activity was malicious, so I would continue reviewing the session rather than overstate the finding.

Analyst Takeaway

The initial traffic confirms successful DNS resolution and TCP connection establishment from the flagged workstation to an external destination. The behavior warrants deeper inspection of the encrypted session and subsequent traffic, but the available evidence is not sufficient to classify the connection as malicious.

Phase 2 — TLS Session Analysis

```
3.465201  192.168.1.45 → 198.51.100.42   TLSv1.3  Client Hello
 3.472991  198.51.100.42 → 192.168.1.45   TLSv1.3  Server Hello
 3.476112  198.51.100.42 → 192.168.1.45   TLSv1.3  Application Data
 3.477620  192.168.1.45 → 198.51.100.42   TLSv1.3  Application Data
```

My Investigation Response

The client and server successfully negotiated a TLS 1.3 session using a Client Hello and Server Hello. Encrypted application data was then exchanged in both directions. Because the payload is encrypted, the packet capture does not reveal the contents of the application data. I would continue by examining traffic volume, direction, timing, and additional connections rather than assuming what the encrypted data contains.

Escalation Note

Summary: Workstation 192.168.1.45 established a TCP connection to 198.51.100.42 over port 443, successfully negotiated TLS 1.3, and exchanged encrypted application data.

Assessment: The connection was technically successful, but encryption prevents direct inspection of the application payload. The session alone is not sufficient to determine whether the communication is benign or malicious.

Recommendation: Continue the network timeline, compare byte and packet direction, review additional destination domains and IPs, and look for repeated or automated communication patterns.

Phase 3 — Repeated Outbound Transfer

```
7.102114  192.168.1.45 → 8.8.8.8        DNS      Standard query A cdn-sync.test
 7.114882  8.8.8.8 → 192.168.1.45        DNS      Response A 203.0.113.88
 7.117330  192.168.1.45 → 203.0.113.88    TCP      52381 → 443 [SYN]
 7.127802  192.168.1.45 → 203.0.113.88    TLSv1.3  Client Hello
 7.139771  192.168.1.45 → 203.0.113.88    TLSv1.3  Application Data (1460 bytes)
 7.141003  192.168.1.45 → 203.0.113.88    TLSv1.3  Application Data (1460 bytes)
```

My Investigation Response

The workstation later resolved cdn-sync.test to 203.0.113.88 and established another encrypted TLS session over port 443. Unlike the earlier session, multiple consecutive application-data packets were sent from the internal workstation to the external destination. Conversation statistics showed 184 packets totaling 238,420 bytes outbound compared with 26 packets totaling 18,991 bytes inbound. The asymmetry is consistent with an outbound data transfer and warrants investigation for possible exfiltration, but traffic volume alone does not prove that sensitive information was transferred.

Escalation Note

Summary: Workstation 192.168.1.45 established an encrypted session with 203.0.113.88 and transmitted substantially more data outbound than it received.

Assessment: The outbound-heavy traffic is consistent with a larger data transfer. Because the payload is encrypted, the packet capture cannot establish what data was sent or whether the transfer was authorized.

Recommendation: Review the complete conversation, validate the destination, identify the initiating process on the endpoint, and determine whether the transfer matches approved application behavior.

Phase 4 — Periodicity and Possible Exfiltration

```
60.102001   192.168.1.45 → 203.0.113.88  TCP      52501 → 443 [SYN]
 120.114882  192.168.1.45 → 203.0.113.88  TCP      52544 → 443 [SYN]
 180.109331  192.168.1.45 → 203.0.113.88  TCP      52602 → 443 [SYN]
 240.121004  192.168.1.45 → 203.0.113.88  TCP      52661 → 443 [SYN]
 Endpoint context: unknown process initiates the connections; no approved
application is documented as using cdn-sync.test.
```

My Investigation Response

The same external connection recurred at approximately 60-second intervals, suggesting automated or beacon-like behavior. Periodicity can be legitimate, such as an update or monitoring service, so the timing does not establish malware by itself. However, the affected workstation was the only observed host contacting cdn-sync.test, no approved application was documented as using the domain, an unknown process initiated the connections, and the sessions remained outbound-heavy. Taken together, I would classify the activity as highly suspicious and escalate for endpoint and threat-intelligence analysis.

Final Escalation Note

Summary: Workstation 192.168.1.45 repeatedly connected to cdn-sync.test / 203.0.113.88 at approximately 60-second intervals and transmitted significantly more data outbound than inbound.

Assessment: The combined evidence is highly suspicious and is consistent with possible automated command-and-control or data-exfiltration behavior. The available packet evidence does not independently prove that sensitive data was transmitted or that the destination is malicious.

Recommendation: Escalate to Tier 2 or incident response, preserve the packet capture and timeline, isolate the workstation if organizational procedures support it, identify the unknown process, validate the destination using approved threat-intelligence sources, and review endpoint/proxy telemetry for evidence of unauthorized file transfer.

Consolidated Investigation Timeline

Time	Evidence	Observed Activity	Interpretation

3.441–3.464	DNS / TCP	DNS resolution and TCP handshake to updates-check.test / 198.51.100.42	Connection established
3.465–3.478	TLS 1.3	Client/Server Hello followed by bidirectional application data	Encrypted session established
7.102–7.142	DNS / TLS	cdn-sync.test resolves and the host sends repeated TLS application-data packets	Outbound transfer pattern
Conversation	Statistics	184 packets / 238,420 bytes outbound vs. 26 packets / 18,991 bytes inbound	Outbound-heavy traffic
~60 sec	Timing	Connections recur at approximately 60-second intervals	Automated / beacon-like pattern
Baseline	TLS 1.3	Only one workstation contacts cdn-sync.test during the capture	Environment-specific anomaly
Software	Inventory	No approved application is documented as using cdn-sync.test	Legitimacy not established
Endpoint	Process	An unknown process initiates the recurring external connections	Raises suspicion
Final	Assessment	Repeated encrypted outbound transfer combined with periodicity and unknown process context	Highly suspicious; escalate

Network Behaviors Demonstrated

DNS and Connection Establishment: The workstation resolved external domains and successfully completed TCP handshakes to destination port 443.

Encrypted Communication: TLS 1.3 negotiation succeeded and application data was exchanged, limiting direct payload inspection from the packet capture.

Outbound Transfer Pattern: Conversation statistics and repeated TLS records showed substantially more traffic leaving the workstation than returning.

Beaconing / Exfiltration Concern: Connections recurred at near-regular intervals and were initiated by an unknown process. The pattern is consistent with possible automated command-and-control or data exfiltration, but additional endpoint and application evidence is required for confirmation.

Skills Demonstrated

- Wireshark display filtering and host traffic isolation.
- DNS query/response and domain-to-IP correlation.
- TCP three-way handshake and connection-state analysis.
- TLS 1.3 session and encrypted application-data interpretation.
- Conversation statistics, byte direction, and traffic-volume analysis.
- Periodicity and beacon-like behavior identification.
- Evidence-based Tier 1 SOC assessment, escalation writing, and investigation handoff.

Final Case Conclusion

The investigation began with a medium-severity alert for unusual outbound traffic and progressed into a highly suspicious network-activity case. Wireshark analysis confirmed DNS resolution, successful TCP connection establishment, TLS 1.3 negotiation, repeated outbound-heavy encrypted transfers, and connections recurring at approximately 60-second intervals. Environmental context strengthened the concern because only one workstation contacted the domain, no approved application was documented as using it, and an unknown process initiated the connections. The evidence is consistent with possible automated command-and-control or data exfiltration behavior, but the encrypted network traffic does not prove that sensitive data was transferred or that the external destination was malicious. The appropriate Tier 1 action is escalation with a preserved packet capture, clear timeline, and recommendations for endpoint, proxy, and threat-intelligence validation.

Portfolio Note

This project is a simulated training lab created to demonstrate SOC analyst network-investigation methodology. All domains, IP addresses, hosts, processes, and events are fictional or reserved for documentation/training purposes and used for educational purposes.